Cross Lingual BERT Adaptation English Spanish

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Problem definition and motivation

Pretrained language models such as BERT have achieved strong performance across a wide range of natural language processing tasks. However, most widely used models, such as bert-base-uncased, are primarily trained on English corpora. When applied to other languages, such as Spanish, their performance often degrades due to mismatches in vocabulary, tokenization, and linguistic structure.

Although multilingual models exist, this work aims to explore the extent to which a pretrained English language model can adapt to a new language with minimal prior exposure. Understanding this capability may provide insight into whether large language models can generalize to low-resource or even previously unseen languages, with potential applications in deciphering ancient or under-documented languages.

A key challenge in working with such languages is the limited availability of training data. To investigate this constraint, we evaluate how effectively BERT can learn from relatively small amounts of Spanish text. To maintain a consistent baseline across experiments, we fix the tokenization scheme and vary only the amount of training data.

We focus on adapting BERT to Spanish using a large-scale Spanish corpus, and evaluate whether such adaptation improves the model's ability to model Spanish text.

Related Work

- Conneau et al. (2020) — XLM-R (https://arxiv.org/abs/1911.02116)

XLM-R is a multilingual masked language model trained on 100 languages that significantly outperforms mBERT on cross-lingual tasks, especially for low-resource languages.

- Artetxe et al. (2020) — Cross-lingual Transferability of Monolingual Representations (https://arxiv.org/abs/1910.11856)

Demonstrate that adapting only the embedding layer via MLM enables cross-lingual transfer, indicating that transformer representations generalize across languages and supporting vocabulary expansion methods.

- Pfeiffer et al. (2020) — MAD-X (https://arxiv.org/abs/2005.00052)

MAD-X enables parameter-efficient cross-lingual transfer using lightweight adapters, providing an alternative to full model fine-tuning.

Dataset description

We use the "120 Million Word Spanish Corpus" from Kaggle as the primary dataset, providing a representation of real-world Spanish language usage. The data is scraped from Wikipedia articles.

The related github repo for all of the code is HERE.

Model Design

To systematically evaluate the impact of continued pretraining and tokenizer adaptation, we design three model variants:

- **1. English baseline model (BERT-base-uncased)**
  - Applied directly to Spanish text without any adaptation
  - Serves as a baseline to measure cross-lingual performance degradation
- **2. Continued pretraining with tokenizer adaptation**
  - Extend the original tokenizer by adding Spanish-specific subword tokens learned from the dataset
  - **Limiting token count and article cap**
  - Initialize embeddings for new tokens and continue MLM training on Spanish data
  - Evaluates the combined effect of model adaptation and improved tokenization
- **3. Multilingual or Spanish-pretrained model**
  - Trained on multilingual from the beginning
  - Serves as an upper-bound reference for cross-lingual performance

Training Procedure

Overall Pipeline

```
Data preprocessing and dataset construction
        ↓
Extract Spanish-specific subwords
        ↓
Expand the BERT vocabulary with selected new tokens
        ↓
Resize embedding and re-tokenize dataset
```

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        ↓
Train and evaluate with masked language modeling (MLM)
```

1. **Data preprocessing and dataset construction**

We start by processing the raw Spanish corpus by combining files, tracking the number of articles. We apply basic cleaning, including removing markup tags, fixing whitespace, and filtering out noisy content. Very short articles are removed to ensure better data quality.

We allocate approximately 80% of the data for training and 20% for validation.

Split	Size
Train	204,984
Validation	51,467

In our experiments we will see what happens when we cut the avaliable data to 100,000 and 50,000 articles.

2. **Extract Spanish-specific subwords**

After preprocessing, we train a SentencePiece tokenizer on the cleaned Spanish training articles to learn a subword vocabulary of size **10,000**. These learned Spanish-specific subwords are used as a candidate pool for expanding the original BERT tokenizer in the next step.

3. **Expand the BERT vocabulary with selected new tokens**

In BERT, tokenization is typically performed using a WordPiece-based subword algorithm, which decomposes each word into a sequence of subword tokens:

$$\text{word} \rightarrow [t_1, t_2, \ldots, t_n], \quad t_i \in \mathcal{V}$$

This limitation becomes apparent when applying an English-pretrained tokenizer to Spanish text. For example, consider the sentence:

> **"Yo estaba aprendiendo español"**

Using the original English BERT tokenizer, the sentence is fragmented as:

> **"[Yo] [est] [##aba] [aprend] [##iendo] [es] [##pañ] [##ol]"**

To address this issue, we expand the tokenizer vocabulary by adding new tokens that better match Spanish morphology. After vocabulary expansion, the same sentence can be represented as:

> **"[Yo] [estaba] [aprendiendo] [español]"**

To study the impact of vocabulary expansion, we experiment with different levels of token addition:

- **Level 1 (Small expansion):** 500 tokens
- **Level 2 (Medium expansion):** 1,000 tokens
- **Level 3 (Large expansion):** 2,000 tokens

For each level, the tokenizer is extended with the corresponding number of tokens.

4. **Resize Embedding and Re-tokenize Dataset**

After expanding the tokenizer, we resize the model's embedding layer to match the updated vocabulary size. The embedding matrix is defined as:

$$E \in \mathbb{R}^{V \times d}, \quad \mathbf{e}_i = E[i]$$

where V is the vocabulary size and d is the embedding dimension. Each token i is mapped to its corresponding vector representation $\mathbf{e}_i$.

With vocabulary expansion, the embedding matrix is resized from:

$$E_{\text{old}} \in \mathbb{R}^{V \times d} \quad \rightarrow \quad E_{\text{new}} \in \mathbb{R}^{(V+k) \times d}$$

where k is the number of newly added tokens. The embeddings for these new tokens are randomly initialized.

5. **Train and evaluate with masked language modeling (MLM)**

This continued pretraining enables the baseline model to adapt from English to Spanish and learn meaningful representations for the newly added tokens. We will adjust the avaliable data (articles) and new token count. Performance is evaluated on the validation set using the selected metrics.

Evaluation Metrics

These metrics capture complementary aspects of model performance, including language modeling quality, tokenization efficiency, predictive accuracy, and training dynamics using the following metrics:

1. **Pseudo-Perplexity (PPL)**
   Measures the model's uncertainty in predicting Spanish text, approximated from the masked language modeling (MLM) loss:

$$\text{PPL} = \exp\left(\frac{1}{N}\sum_{i=1}^{N} -\log P(x_i \mid x_{\setminus i})\right)$$

where $x_{\setminus i}$ denotes the input sequence with the i-th token masked, and N is the number of tokens. Lower values indicate better language modeling capability.

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2. **Tokenization Efficiency**
   Evaluates how efficiently the tokenizer represents Spanish text:

$$\text{Token Ratio} = \frac{\text{Number of wordpieces}}{\text{Number of words}}$$

A lower ratio indicates fewer subword fragments and better alignment with Spanish morphology.

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3. **Masked Token Prediction Accuracy**
   Measures the proportion of correctly predicted masked tokens:

$$\text{Accuracy} = \frac{\text{Number of correctly predicted masked tokens}}{\text{Total masked tokens}}$$

This provides an interpretable measure of the model's prediction performance.

Experimental results

Table 1. Effect of Vocabulary Expansion Size

Added Tokens	Loss	PPL	Token Ratio	Masked Acc
500	**1.490**	**4.438**	2.704	0.075
1000	1.731	5.644	2.374	0.158
2000	2.047	7.742	**2.046**	**0.225**

Table 1 shows the effect of different vocabulary expansion sizes under Model 2 (BERT with vocab expansion and MLM). Increasing the number of added tokens improves tokenization efficiency and masked token accuracy, but leads to higher loss and perplexity due to insufficiently trained embeddings.

Table 2. Model Performance Comparison

Model	Description	Loss	PPL	Token Ratio	Masked Acc
1	BERT (English baseline)	1.515	5.828	2.876	0.132

2	BERT + vocab expansion + MLM (n = 2000)	2.047	7.742	2.046	0.225
3	Multilingual / Spanish-pretrained	1.251	4.264	2.001	0.402

Table 2 shows that the English BERT baseline achieves lower loss and PPL than Model 2, as it does not introduce randomly initialized tokens and benefits from a fully pretrained embedding space.

Although the loss is low, the masked accuracy is in model 1 is much lower than model 2, indicating that, the baseline fails to recognize other linguistical patterns.

Meanwhile, the multilingual model (Model 3) serves as an upper bound, demonstrating the performance achievable with well-trained language-specific representations.

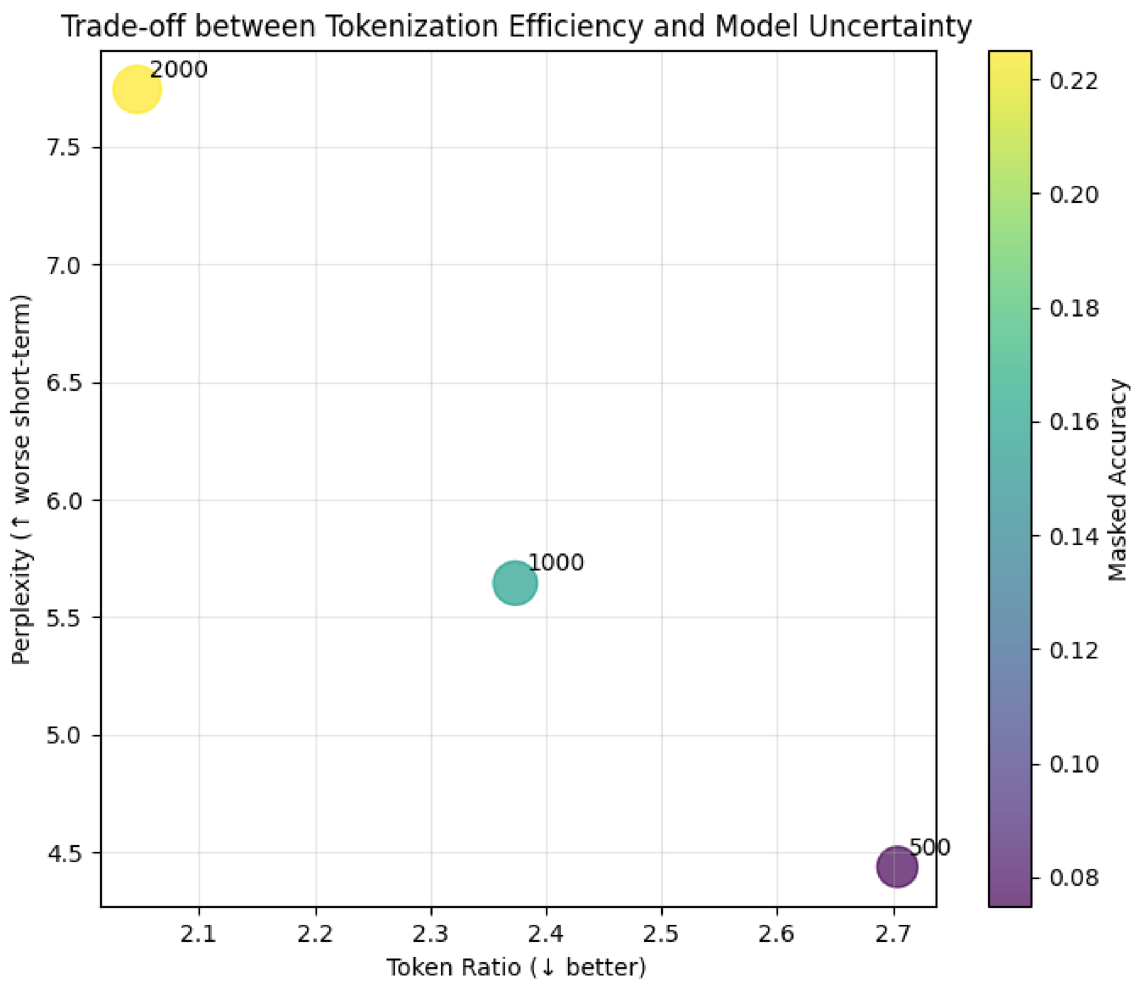
Table 3. Masked Accuracy Across Dataset and Expansion Size

New Tokens ↓ / Articles →	50,000	100,000	251,422(full dataset)
500	0.075	0.075	0.075
1000	0.175	0.175	0.125
2000	0.250	0.200	0.225

Table 3 shows the impact of vocabulary expansion and dataset size on masked token prediction accuracy.

We observe that vocabulary expansion has a stronger effect on accuracy than dataset size, especially when the number of training articles is limited. This suggests that improving tokenization quality is more effective than simply increasing the amount of training data, particularly in low-resource settings.

Trade-off between Tokenization Efficiency and Perplexity



This figure shows that increasing vocabulary size improves tokenization efficiency (lower token ratio) and masked accuracy, **but leads to higher perplexity due to insufficiently trained embeddings**. This demonstrates a short-term vs long-term trade-off in vocabulary expansion.

Analysis and Discussion

Perplexity increases with larger vocabulary expansion

As the number of added tokens increases (e.g., 500 → 2000), the perplexity (PPL) consistently becomes higher.

Explanation:
These embeddings have not yet learned meaningful representations and therefore increase prediction uncertainty during masked language modeling. In addition, a larger vocabulary increases the difficulty of the prediction task, as the model must choose from more possible tokens.

Tokenization efficiency improves

With larger vocabulary expansion, the token ratio decreases

Explanation:
The expanded tokenizer is able to represent Spanish words using fewer subword fragments, leading to more compact and linguistically meaningful representations.

Masked token prediction accuracy improves

Masked token prediction accuracy shows an overall increasing trend as the vocabulary size grows.

Explanation:
Expanded vocabulary allows the model to capture more complete semantic units, making it easier to predict masked tokens correctly.

Limited Benefit from Increasing Dataset Size

Masked token prediction accuracy is more sensitive to vocabulary size than to dataset size, especially when the number of training articles is limited.

Explanation:
Without sufficient vocabulary adaptation, the model still relies on fragmented subword units, limiting its ability to benefit from additional data.

Short-term vs Long-term Effects of Vocabulary Expansion

Aspect	Short-term Effect	Long-term Effect
Embedding Quality	Newly added embeddings are randomly initialized and poorly trained	Embeddings become well-optimized and capture meaningful semantics
Perplexity (PPL)	Increases due to higher uncertainty and untrained embeddings	Expected to decrease as representations improve
Tokenization Efficiency	Improves immediately	Remains stable and continues to benefit model performance
Masked Token Accuracy	Slight improvement	Improves as the model better utilizes refined representations

Overall, vocabulary expansion introduces short-term challenges but provides long-term benefits when sufficient training data and compute are available.

With the results here we are not optimistic that these models would be a good resource for langugage learning. Especially with the limitations put on these models, low vocabulary and low training data, these models performed sometimes performed worse than the base model.

Limitations and Future Work

Due to the scale of the project we were limited to how much we could train each experiment. Here we only trained over one epoch which for these models may be premature, however, we hoped to at least see the behavior.

In the future, we can focus on scaling both the dataset and training process. Using a larger and more diverse Spanish corpus, along with longer pretraining on the BERT model, is expected to further improve language modeling performance and masked token prediction accuracy. Exploring larger or adaptive vocabulary expansion strategies may also better capture linguistic structure and reduce token fragmentation. Additionally, evaluating the adapted model on downstream tasks such as text classification or named entity recognition would provide a more comprehensive assessment of its effectiveness.

Furthermore, an interesting direction would be extend to languages such as Chinese, which have fundamentally different linguistic structures and tokenization characteristics. This would allow us to better understand the effectiveness and limitations of vocabulary expansion in more challenging cross-lingual settings.